

Course: B.Tech

Branch : Common to all Branches

Subject Code & Name: 24AF1000ES206B - PROGRAMMING FOR PROBLEM SOLVING

Semester :II

Max Marks: 60

Date: 19/01/2026

Duration:

3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

			(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)			12
1	Which of the following is considered the brain of the computer? a) RAM b) Hard Disk c) CPU d) Monitor			1
2	Which data type is used to store a single character in C? a) int b) char c) float d) double			1
3	Which of the following is an input device? a) Printer b) Monitor c) Keyboard d) Plotter			1
4	Which operator is used for logical AND? a) & b) && c) d)			1
5	Which keyword is used to exit a loop immediately? a) continue b) break c) exit d) return			1
6	Which of the following is a user-defined function? a) printf() b) scanf() c) main()			1

	d) factorial()		
7	Which symbol is used to access array elements? a) () b) {} c) [] d) <>		1
8	Character arrays are used to store: a) Integers b) Characters c) Strings d) Float values		1
9	Which pointer is used to point to an array? a) char pointer b) void pointer c) array pointer d) pointer to array		1
10	Which keyword is used to define a structure? a) struct b) structure c) typedef d) union		1
11	Which function is used to open a file? a) open() b) fopen() c) fileopen() d) read()		1
12	EOF stands for: a) End of File b) End of Function c) Error on File d) Exit of File		1
Q. 2	Solve the following.		12
A)	Describe the steps involved in program development. Explain compilation errors and runtime errors.		6
B)	Describe the basic anatomy of a computer system with a neat diagram.		6
Q.3	Solve the following.		12
A)	Write a C program to check whether a number is even or odd.		6
B)	Explain arithmetic, relational, and logical operators with examples.		6
Q. 4	Solve Any Two of the following.		12
A)	Explain functions and program structure in C.		6

B)	Write a C program using functions to find the factorial of a number.		6
C)	Explain operator precedence and order of evaluation in detail.		6
Q.5	Solve Any Two of the following.		12
A)	Write a C program to read n numbers into an array and find the largest element using pointers.		6
B)	Explain multidimensional arrays with an example and write a C program to perform matrix addition		6
C)	Explain the concept of a pointer to an array with an example.		6
Q. 6	Solve Any Two of the following.		12
A)	Write a C program to write text to a file and read it back.		6
B)	Explain structures and file handling in detail.		6
C)	Explain control flow statements in C: a) if–else ladder b) switch–case		6
	*** End ***		